YUZHUO XIAO

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EDUCATION

GuiZhou University, China (GZU) *June 2026*
B.E. in Computer Science (Advanced Class)

RESEARCH INTERESTS

My research interests focus on Natural Language Processing (NLP), Large Language Models (LLMs), LLM-based Agents, Multi-Agent Systems, and Collective Intelligence.

PUBLICATIONS

[1] Y. Xiao, Z. Han, Y. Wang, and H. Jiang, *Xfacta: Contemporary, real-world dataset and evaluation for multimodal misinformation detection with multimodal llms*, 2025. arXiv: 2508.09999 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2508.09999.

RESEARCH EXPERIENCE

XFacta: Contemporary, Real-World Dataset and Evaluation for Multimodal Misinformation Detection with Multimodal LLMs [1]
Yuzhuo Xiao*, *Zeyu Han**, *Yuhan Wang, Huaizu Jiang* NEU (remote)

- Project leader
- Project: XFACTA, a contemporary real-world multimodal misinformation detection project, focusing on dataset construction and systematic analysis of evidence retrieval and reasoning in MLLMs.
- Contributions: Built a large-scale contemporary real-world multimodal misinformation dataset and performed systematic analysis of evidence retrieval and reasoning strategies for MLLM-based detection.
- Outcomes: Achieved state-of-the-art performance on the benchmark and enabled continuous dataset growth via a validated detector-in-the-loop pipeline.

ACADEMIC TRAINING AND SUMMER PROGRAMS

AI Study Tour – University of Newcastle, Australia Spring 2026

- Project: Participated in an AI-focused academic study tour, developed an interactive 3D simulation and visualization system for sea ice and ocean wave flexural-gravity wave dynamics as a major course project.
- Contributions: Led core implementation using Python, PyQt5, NumPy, SciPy, and Matplotlib; integrated wave simulation, 3D visualization, and interactive physical parameter controls; implemented real-time parameter adjustment, animation playback, and visual representations of wave propagation and ice-shelf deformation.
- Outcomes: Developed and presented a functional interactive 3D simulation application, demonstrating the integration of scientific computing, physical modeling, and interactive visualization.

Multimodal AI Summer Course - Guest Lectures by USTB Summer 2024
Instructors: Dr. Dandan Song (USTB) and Prof. Ruizhang Huang

- Project: Participated in an summer course on multimodal learning, covering vision-language model architectures, multimodal alignment, and representative multimodal frameworks.

- Contributions: Studied multimodal learning architectures and frameworks; completed a project involving the deployment and adaptation of MiniCPM, a Multimodal LLM, for downstream tasks.
- Outcomes: Gaining practical experience in multimodal model deployment, adaptation, and downstream applications.

AI & Robotics Training Program – SenseTime, Shang Hai Summer 2023

- Project: Participated in a training program on deep learning, computer vision, and embodied perception.
- Contributions: Built a custom visual dataset, trained image classification models, and integrated vision sensors for gesture- and pose-based action recognition.
- Outcomes: Gained hands-on experience in computer vision model development and vision-driven robotic interaction.

TECHNICAL EXPERIENCE

RecSys Challenge 2025: Universal User Behavior Representation Competition Project

- Project: Universal behavioral profile learning for multi-task recommendation and user modeling.
- Contributions: Manually designed and implemented behavior-based statistical features from large-scale user interaction logs; explored feature construction strategies focusing on behavior frequency, temporal dynamics, and cross-action relationships; conducted systematic ablation studies to analyze the impact of feature quality versus model complexity.
- Outcomes: Built a robust and reproducible feature-driven user representation pipeline that consistently improved performance across multiple public and hidden downstream tasks under changing data distributions.

Parameter-efficient Fine-tuning for Machine Translation Course Project

- Project: An end-to-end machine translation project for low-resource Bengali–English translation using parameter-efficient fine-tuning.
- Contributions: Designed and implemented a LoRA-based fine-tuning pipeline on mBART-large; performed data cleaning, filtering, and sampling on large-scale parallel corpora; trained models on 300K sentence pairs.
- Outcomes: Achieved over 200% relative BLEU improvement over the pretrained mBART baseline on held-out news test sets, with consistent gains in ChrF.

Digital Image Processing: Classical Methods and Deep Learning Course Project

- Project:Classical and Deep Learning–Based Image Processing System
- Contributions: Implemented edge detection, frequency-domain processing, color space conversion, image segmentation, and object detection algorithms. Applied U-Net for pixel-level defect segmentation and compared with traditional methods.
- Outcomes:Built end-to-end image processing pipelines using OpenCV and PyTorch.

AWARDS AND HONORS

DaBeiNong First-Class Scholarship *Summer 2025*
Guizhou University First-Class Scholarship *Summer 2023*
Guizhou University "Three Good Student" Award *Summer 2023*
Guizhou University "Outstanding Graduate" Award *Summer 2026*

SKILLS/HOBBIES

Programming Languages Python, C/C++, Bash
Machine Learning Tools Pytorch, Tensorflow, Sklearn, Pandas, Numpy, OpenCV
Hobbies badminton and dancing